# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The number of perfect squares between $100$ and $1000$ (both excluded) is: 1

**(a)** $20$ **(b)** $22$ **(c)** $21$ **(d)** $23$

**2.** For the expression $n^2 - n + 11$, the smallest natural number $n$ for which the value is NOT prime is: 1

**(a)** $12$ **(b)** $10$ **(c)** $13$ **(d)** $11$

**3.** For the expression $n^2 + n + 191$, the smallest natural number $n$ for which the value is NOT prime is: 1

**(a)** $3$ **(b)** $2$ **(c)** $5$ **(d)** $4$

**4.** The sum of 5 consecutive integers is $180$. The largest of them is: 1

**(a)** $38$ **(b)** $40$ **(c)** $39$ **(d)** $37$

**5.** If $n$ is any integer not divisible by 3, then $n^2$ leaves a remainder ___ on division by 3. For $7$ this gives: 1

**(a)** $3$ **(b)** $1$ **(c)** $0$ **(d)** $2$

**6.** For the two-digit number $83$, the value of (number − digit sum) ÷ 9 is: 1

**(a)** $8$ **(b)** $10$ **(c)** $9$ **(d)** $7$

**7.** If the sides of a triangle are $9$, $40$ and $x$ (with $x$ the longest side) and it is right-angled, then $x$ is: 1

**(a)** $43$ **(b)** $42$ **(c)** $41$ **(d)** $40$

**8.** The smallest natural number with exactly 6 factors (counting 1 and itself) is: 1

**(a)** $10$ **(b)** $13$ **(c)** $12$ **(d)** $16$

**9.** **Assertion (A):** The converse of 'If a number is divisible by 6, then it is even' is false. 1
**Reason (R):** The number 4 is even but is not divisible by 6.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** The statement 'Every prime number is odd' is false. 1
**Reason (R):** The prime number 2 is not odd.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Use the converse of the Baudhāyana–Pythagoras theorem: for what value of $x$ is a triangle with sides $20, 21, x$ ($x$ the longest) right-angled? 2

**12.** Show that "If a number is divisible by 4 and by 6, then it is divisible by 24" is false. Give a counterexample. 2

**13.** Verify that $n^2 - 1$ is a multiple of 8 for the odd number $n = 11$. Write the converse of "If $n$ is odd, then $n^2 - 1$ is a multiple of 8". 2

**14.** Use the converse of the Baudhāyana–Pythagoras theorem: for what value of $x$ is a triangle with sides $11, 60, x$ ($x$ the longest) right-angled? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Prove that the sum of the squares of two consecutive integers is odd. Verify it for $8$ and $9$. 3

**16.** Prove: if 3 divides $x$ and $x$ divides $y$, then 3 divides $y$. Verify with $x = 12$ and $y = 60$. 3

**17.** For $m = 5$ and $n = 2$ show that $(21,\ 20,\ 29)$ are the sides of a right triangle, using the converse of the Baudhāyana–Pythagoras theorem. 3

OR

Prove that the sum of the squares of two consecutive integers is odd. Verify it for $12$ and $13$. 3

**18.** Prove: if 7 divides $x$ and $x$ divides $y$, then 7 divides $y$. Verify with $x = 14$ and $y = 84$. 3

OR

For $m = 6$ and $n = 1$ show that $(35,\ 12,\ 37)$ are the sides of a right triangle, using the converse of the Baudhāyana–Pythagoras theorem. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** In a maths club, Naveen considers the statement: 'If a number is a multiple of 8, then it is a multiple of 4.' 5
(i) Identify the hypothesis and the conclusion.
(ii) Write the converse of the statement.
(iii) Is the original statement true?
(iv) Is the converse true? Support your answer with a counterexample if it is false.

**20.** Prove that the product of three consecutive integers is divisible by 6. Verify it for $9, 10, 11$. 5

OR

Prove that the square of any odd integer is $1$ more than a multiple of $8$. Verify it for $13$. 5

## END OF QUESTION PAPER — Total: 40 Marks